

pin 7 of IC3 goes high and switches on LED2 and conducts transistor T2, and RL1 and RL2 energise.

Output voltages of IC1 and IC2 are regulated by VR2 and are available for charging at CON4. Once the battery under charging attains its full charging voltage, pin 1 of IC3 goes high, as indicated by the glowing of LED4, and T1 conducts, which reduces the voltage at pin 1 of IC1 and IC2.

When a 12V battery is connected in reverse polarity, D11 conducts, which, in turn, switches on the piezo buzzer and LED5 glows. This protection is important while charging batteries externally.

Construction and testing

An actual-size, single-side PCB for the 12V battery charger-cum-variable power supply is shown in Fig. 2 and its component layout in Fig. 3.

After assembling the circuit on the PCB, place it in a suitable cabinet. Fit CON3 and CON4 on the front side of the cabinet. CON3 is for variable power supply (VPS*) and CON4 for battery under charging (BUC#).

Connect CON3 between N/C2 terminal of relay RL2 and point 4 given on the PCB. Connect CON4 between points 3 and 5 given on the PCB. Fix VR1 through VR5 and all LEDs on front side of the cabinet, and label these as shown in the circuit diagram.

Place transformer X1 inside the cabinet. Fix CON1 and switch S1 on the rear side of the cabinet. Use proper heat-sinks for IC1, IC2 and IC4, and connect a 12V cooling fan at CON2 for fast cooling.

The circuit works off 230V mains power supply. Refer the test points table for checking the voltages at various points before using the circuit.

Initial setup and testing

1. Remove jumper J1 and J3, connect J2 and switch on S1.
2. Adjust VR1 and VR2 to get 9V DC (as battery dead voltage or as

PARTS LIST

Semiconductors:	
IC1, IC2	- LM138, variable voltage regulator
IC3	- LM338 dual op-amp
IC4	- 7812, 12V voltage regulator
T1	- BC547 npn transistor
T2	- 2N2222 npn transistor
LED1-LED7	- 5mm LED
D1-D5	- 6A4 rectifier diode
D6-D13	- 1N4007 rectifier diode
ZD1	- 4.7V zener diode
Resistors (all 1/4-watt, 5% carbon, unless stated otherwise):	
R1	- 270-ohm
R2, R4-R7, R9, R11	- 3.3-kilo-ohm
R3	- 0.1-ohm, 5Watt
R8	- 10-kilo-ohm
R10	- 100-ohm
VR1, VR2, VR4, VR5	- 5-kilo-ohm potmeter
VR3	- 47-kilo-ohm potmeter
Capacitors:	
C1	- 4700µF, 40V electrolytic
C2, C4	- 10µF, 25V electrolytic
C3	- 100µF, 35V electrolytic
C5	- 0.1µF ceramic disk
Miscellaneous:	
X1	- 230V AC primary to 15V-0-15V, 5A secondary transformer
RL1	- 12V, 1C/O PCB-mounted relay
RL2	- 12V, 1C/O, 10A contact-current rating relay
CON1, CON2	- 2-pin connector
CON3, CON4	- 2-pin connector terminal for 10A
S1	- On/off switch (6A)
PZ1	- Piezo buzzer
J1-J3	- 2-pin shorting jumper connector
	- Heat-sink for IC1, IC2, IC4
	- 12V cooling fan

required) with respect to ground at TP6.

3. Adjust VR4 such that you can turn on and off status LED2 and LED6 alternately. RL1 and RL2 also change their states (energise/de-energise).
4. Adjust VR2 to get the full battery voltage (say, 13.4V DC with respect to ground) at TP6.
5. Adjust VR5 to turn on LED4, if TP7 is connected to ground through J3.
6. Once LED4 is set, adjust VR2 to get battery charging voltage (say, 14.2V DC with respect to ground) at TP6.
7. Connect J1 (J2 still connected) and adjust VR3 to get battery standby voltage (say, 13.4V DC

Test Points

Test point	Details
TP0	0V (GND)
TP1	Around 20V DC when S1 is closed
TP2	1.2V to 18V
TP3	1.2V to 18V
TP4	0.5V to 18V
TP5	0.5V to 18V
TP6	Around battery voltage
TP7	0 when J3 is grounded or 12V when J1 is not grounded

with respect to ground) at TP6.

8. Once the required voltages are set, remove J2. The circuit is ready to use.
 9. If no battery is connected at CON4, variable voltage is obtained at CON3 by using VR1. Brightness of LED7 is proportional to the output voltage at CON3.
 10. If 12V ($\geq 50\text{Ah}$) battery is connected at CON4 in reverse polarity, buzzer PZ1 gives an alarm sound and LED5 glows.
 11. If the same battery is connected in correct polarity at CON4, status of the battery is indicated by LED2 (healthy) or LED6 (dead). RL1 and RL2 will get energised in case the battery is healthy and the battery will get charging voltage. Once the battery is fully charged, LED4 glows, which indicates that the battery is fully charged and the charger is at standby voltage.
 12. Use proper crocodile clips for connecting the battery terminals.
- EFY note:** 1. Charging voltage, standby voltage and battery dead voltage may vary as per the manufacturer of the battery. The values indicated here are safe voltage values.
2. Heat-sink with mica insulation must be provided for IC1 and IC2. **EFY**



Fayaz Hassan is a manager at Visakhapatnam Steel Plant, Visakhapatnam, Andhra Pradesh. He is interested in microcontroller projects, mechatronics and robotics